HW0512&0519

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15.6 Using the NLS panel data on $N = 716$ young women, we consider only years 1987 and 1988. We are interested in the relationship between ln(*WAGE*) and experience, its square, and indicator variables for living in the south and union membership. Some estimation results are in Table 15.10.

TABLE 15.10 **Estimation Results for Exercise 15.6**

	(1) OLS 1987	(2) OLS 1988	(3) FE	(4) FE Robust	(5) RE
C	0.9348	0.8993	1.5468	1.5468	1.1497
	(0.2010)	(0.2407)	(0.2522)	(0.2688)	(0.1597)
EXPER	0.1270	0.1265	0.0575	0.0575	0.0986
	(0.0295)	(0.0323)	(0.0330)	(0.0328)	(0.0220)
$EXPER^2$	−0.0033	−0.0031	−0.0012	−0.0012	−0.0023
	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0007)
SOUTH	−0.2128	−0.2384	−0.3261	−0.3261	−0.2326
	(0.0338)	(0.0344)	(0.1258)	(0.2495)	(0.0317)
UNION	0.1445	0.1102	0.0822	0.0822	0.1027
	(0.0382)	(0.0387)	(0.0312)	(0.0367)	(0.0245)
N	716	716	1432	1432	1432

a. The OLS estimates of the ln(*WAGE*) model for each of the years 1987 and 1988 are reported in columns (1) and (2). How do the results compare? For these individual year estimations, what are you assuming about the regression parameter values across individuals (heterogeneity)?

(a)

1987 年與 1988 年 OLS 估計結果差異非常小 OLS 模型假設全體個體於橫斷面的參數值（包括截距）都是相同的，不存在個體之間的異質性。

b. The ln(*WAGE*) equation specified as a panel data regression model is

$$\ln(WAGE_{it}) = \beta_1 + \beta_2 EXPER_{it} + \beta_3 EXPER_{it}^2 + \beta_4 SOUTH_{it} + \beta_5 UNION_{it} + (u_i + e_{it}) \qquad \text{(XR15.6)}$$

Explain any differences in assumptions between this model and the models in part (a).

(b)

該模型明確地將誤差項分解為 ui（隨個體改變、不隨時間改變的未觀察異質性，）與 eit（隨時間與個體改變的隨機誤差）。

該模型假設各變數的真實參數（β1…β5）在 1987 年與 1988 年之間是固定不變的；而(a)的模型允許各年份有不同的參數。

c. Column (3) contains the estimated fixed effects model specified in part (b). Compare these estimates with the OLS estimates. Which coefficients, apart from the intercepts, show the most difference?

(c)

係數對比：

EXPER：由 OLS 的 0.127 大幅下降至 FE 的 0.0575。

SOUTH：由 OLS 的−0.24 變成明顯更負的−0.3261。

差異最大的係數：

除了截距項外， EXPER 及 SOUTH 的係數變化最為劇烈。

d. The F-statistic for the null hypothesis that there are no individual differences, equation (15.20), is 11.68. What are the degrees of freedom of the F-distribution if the null hypothesis (15.19) is true? What is the 1% level of significance critical value for the test? What do you conclude about the null hypothesis.

(d)

F 分布的**分子自由度**為 N−1=716−1=715。**分母自由度**為

NT−N−(K−1)=1432−716−4=712。

其中 N=716 是個體數量，T=2 是時間點數量，K=5 是模型中參數個數。1%顯著水準下的臨界值約為 1.19，因為 F=11.68 大於臨界值，故拒絕虛無假設，表示存在顯著的未觀察的個體差異(ui≠0)。。

e. Column (4) contains the fixed effects estimates with cluster-robust standard errors. In the context of this sample, explain the different assumptions you are making when you estimate with and without cluster-robust standard errors. Compare the standard errors with those in column (3). Which ones are substantially different? Are the robust ones larger or smaller?

(e)

假設上的差異：

不使用穩健標準誤：假設隨機誤差項 eit 服從獨立同分佈（i.i.d.），即滿足同質變異性（Homoskedasticity），且同一個體在不同期別之間的誤差項不存在任何序列相關。

使用群集穩健標準誤：放寬上述限制，允許同一個體（群集）內部的誤差項存在任意形式的異質變異與跨期序列相關性，僅要求不同個體之間保持獨立。

標準誤比較：

對比第(3)欄與第(4)欄，SOUTH 的標準誤差異最為巨大，從 0.1258 幾近翻倍膨脹至 0.2495。

大小與結論：

整體而言，穩健標準誤明顯較大（特別是 SOUTH）。

f. Column (5) contains the random effects estimates. Which coefficients, apart from the intercepts, show the most difference from the fixed effects estimates? Use the Hausman test statistic (15.36) to test whether there are significant differences between the random effects estimates and the fixed effects estimates in column (3) (Why that one?). Based on the test results, is random effects estimation in this model appropriate?

(f)

與 FE 差異最大的係數：

除了截距外，EXPER（FE: 0.0575 vs RE: 0.0986）與 SOUTH（FE: −0.3261 vs RE: −0.2326）的係數差異最大。

Hausman t 統計量：

	t-value
EXPER	-1.67
$EXPER^2$	1.29
SOUTH	-0.77
UNION	-1.06

為什麼豪斯曼檢定要與第(3)欄進行對比？

傳統的豪斯曼檢定基於一個核心前提：在虛無假設（H0: no endogeneity）成立時，隨機效果（RE）估計量必須是漸進有效率（Asymptotically Efficient）的。而此效率性建立在古典 i.i.d.誤差項假設之上。第(3)欄的傳統 FE 也是在

完全相同的 i.i.d.假設下估計的。此時，兩者變異數差的簡化公式才會成立。

隨機效果模型在此是否合適？

只有 **EXPER** 係數在 10%的顯著水準下顯示有顯著差異。因此內生性的證據較弱，無法根據這些結果強烈反對使用隨機效果估計量。故使用隨機效果模型是適當的。

15.20 This exercise uses data from the STAR experiment introduced to illustrate fixed and random effects for grouped data. In the STAR experiment, children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star*.

a. Estimate a regression equation (with no fixed or random effects) where *READSCORE* is related to *SMALL*, *AIDE*, *TCHEXPER*, *BOY*, *WHITE_ASIAN*, and *FREELUNCH*. Discuss the results. Do students perform better in reading when they are in small classes? Does a teacher's aide improve scores? Do the students of more experienced teachers score higher on reading tests? Does the student's sex or race make a difference?

b. Reestimate the model in part (a) with school fixed effects. Compare the results with those in part (a). Have any of your conclusions changed? [*Hint*: specify *SCHID* as the cross-section identifier and *ID* as the "time" identifier.]

c. Test for the significance of the school fixed effects. Under what conditions would we expect the inclusion of significant fixed effects to have little influence on the coefficient estimates of the remaining variables?

d. Reestimate the model in part (a) with school random effects. Compare the results with those from parts (a) and (b). Are there any variables in the equation that might be correlated with the school effects? Use the LM test for the presence of random effects.

e. Using the *t*-test statistic in equation (15.36) and a 5% significance level, test whether there are any significant differences between the fixed effects and random effects estimates of the coefficients on *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*. What are the implications of the test outcomes? What happens if we apply the test to the fixed and random effects estimates of the coefficient on *BOY*?

f. Create school-averages of the variables and carry out the Mundlak test for correlation between them and the unobserved heterogeneity.

(a)

$$READSCORE = \beta_0 + \beta_1 SMALL + \beta_2 AIDE + \beta_3 TCHEXPER + \beta_4 BOY + \beta_5 WHITE\_ASIAN + \beta_6 FREELUNCH$$

Table: Pooled OLS model for STAR Experiment

term	estimate	std.error	statistic	p.value
(Intercept)	437.764	1.346	325.180	0.000
small	5.823	0.989	5.886	0.000
aide	0.818	0.953	0.858	0.391
tchexper	0.492	0.070	7.080	0.000
boy	-6.156	0.796	-7.733	0.000
white_asian	3.906	0.954	4.096	0.000
freelunch	-14.771	0.890	-16.592	0.000

(i) 若學生在小班而非大班學習，其平均閱讀成績將高出 5.8 分。

(ii) 是否有助理老師對平均閱讀成績沒有顯著影響。

(iii) 每增加一年教學經驗，平均閱讀成績預估提升 0.49 分。

(iv) 男生的平均閱讀成績比女生低 6 分，白人或亞裔學生的平均閱讀成績則比黑人學生高 3.9 分。

除了 AIDE 變數外，其他所有變數的係數皆達顯著水準。

(b)

重新估計(a)題的模型，但加入「學校固定效果（school fixed effects）」，控制個體之間不變的特質。

Table: School Fixed Effects using 'within' model

term	estimate	std.error	statistic	p.value
small	6.49023	0.91296	7.10898	0.00000
aide	0.99609	0.88169	1.12974	0.25863
tchexper	0.28557	0.07084	4.03090	0.00006
boy	-5.45594	0.72759	-7.49865	0.00000
white_asian	8.02802	1.53566	5.22775	0.00000
freelunch	-14.59357	0.88001	-16.58348	0.00000

(i) 小班教學使平均閱讀分數提高 6.49 分，比 OLS 模型所估計的略高。

(ii) 教學經驗對平均閱讀分數的估計影響降為每增加一年經驗，分數提高 0.29 分。

(iii) 男生與女生平均閱讀分數的差異估計比 OLS 結果略小，為-5.456。

(iv) 白人或亞裔學生與黑人學生的平均分數差異大約翻倍，達到 8 分左右。

除了 AIDE 變數外，其他所有變數的係數皆達顯著水準。

(c)

Table: Fixed effects test: Ho:'No school fixed effects'

df1	df2	statistic	p.value	method	alternative
78	5681	16.69783	0	F test for individual effects	significant effects

用 **F 檢定**比較含有學校固定效果模型與不含學校固定效果的模型。如果學校虛擬變數與其他解釋變數之間不相關，那麼將其納入或排除對迴歸係數估計的影響將很小。F 檢定統計量為 16.70，而臨界值 $F_{(0.95,78,5681)}=1.2798$。因此拒絕虛無假設。

(d)

使用隨機效果模型。如下頁圖表。

OLS 模型假設所有個體是同質的，忽略了個體之間可能存在的異質性或不可觀察因素。而相較之下，**隨機效果模型**允許存在個體間不可觀察的隨機異質性，且假設這些異質性與自變數不相關。

(i) 在 STAR 計畫中，學生是在同一學校內隨機分配的，但不是跨學校分配。部分學校位於富裕的學區，部分學校位於較貧困的學區。家庭收入會影響家庭的學習環境和資源，這在一定程度上由「免費午餐（FREELUNCH）」指標所反映。此外，富裕的學區可能支付更高的薪水，且擁有能力更強和經驗更豐富的教師。

(ii) LM 統計量為 81.72，遠大於標準常態分布的臨界值 1.96。因此，有證據顯示學校間存在未觀察的異質性。

Table: The random effects results for the STAR equation

term	estimate	std.error	statistic	p.value
(Intercept)	436.1268	2.0648	211.2217	0.0000
small	6.4587	0.9125	7.0777	0.0000
aide	0.9921	0.8812	1.1260	0.2602
tchexper	0.3027	0.0703	4.3060	0.0000
boy	-5.5121	0.7276	-7.5753	0.0000
white_asian	7.3505	1.4314	5.1353	0.0000
freelunch	-14.5843	0.8747	-16.6740	0.0000

Table: A random effects test for the STAR school equation

statistic	p.value	method	alternative
81.71546	0	Lagrange Multiplier Test - (Honda)	significant effects

(e)

檢驗固定效果（FE）與隨機效果（RE）模型中各個變數係數是否有顯著差異。在 5% significance level 下，t 值都沒有大於臨界值 1.96，都不顯著，代表固定效果（FE）與隨機效果（RE）模型估計的係數沒有統計上顯著差異。

Table: Individual Hausman t-tests (Equation 15.36) for STAR Experiment

Variable	b_FE	b_RE	se_FE	se_RE	t_stat	p_value
small	6.4902	6.4587	0.9130	0.9125	1.1460	0.2518
aide	0.9961	0.9921	0.8817	0.8812	0.1284	0.8978
tchexper	0.2856	0.3027	0.0708	0.0703	-1.9377	0.0527
white_asian	8.0280	7.3505	1.5357	1.4314	1.2181	0.2232
freelunch	-14.5936	-14.5843	0.8800	0.8747	-0.0956	0.9239
boy	-5.4559	-5.5121	0.7276	0.7276	NaN	NaN

當我們將檢定套用至 boy 時，由於該變數在學校之間的分布極其均勻且隨機，隨機效果模型相較於固定效果模型沒有帶來任何效率增益，導致兩者的標準誤完全相同（皆為 0.7276）。這使得公式分母（變異數之差）變為 0，導致 t 檢定產生 NaN。

(f)

Table: Mundlak Robust Wald Test Results (df = 6)

term	null.value	estimate	std.error	statistic	p.value	df.residual	df
bar_small	0	-23.1031986	20.6572959	13.64879	0.0338151	5753	6
bar_aide	0	9.1871643	18.7114556	13.64879	0.0338151	5753	6
bar_tchexper	0	0.8017333	0.5642115	13.64879	0.0338151	5753	6
bar_boy	0	-50.2118999	23.5933289	13.64879	0.0338151	5753	6
bar_white_asian	0	-7.2214127	5.2800874	13.64879	0.0338151	5753	6
bar_freelunch	0	-4.1164425	7.1023918	13.64879	0.0338151	5753	6

虛無假設（H0）：所有學校平均變數（small_avg、aide_avg、tchexper_avg、boy_avg、white_asian_avg、freelunch_avg）的係數同時等於 0。

對立假設（H1）：至少有一個學校平均變數的係數不為 0。

p-value = 0.0338 < 0.05，**拒絕虛無假設**。表示學校平均變數的係數**不全是 0**，即這些學校平均特徵對解釋變數（讀寫分數）有顯著影響。Mundlak 測試結果顯示學校間平均差異是顯著的，代表個體特有的未觀察異質性和解釋變數之間有相關性，所以不適合使用隨機效果模型，要使用固定效果模型來控制這些未觀察到的異質性。